

Project Log: Synthetic Data Generation

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Research Question

Does LLM-generated synthetic tabular data help or hurt classical ML models compared to real data and traditional methods like SMOTE, across varying data availability and mixing ratios?

Project Overview

We generate synthetic tabular data using Claude Sonnet 4.6 for three UCI datasets, then train classical ML models (logistic regression, SVM, XGBoost, MLP) on different mixtures of real and synthetic data. We compare LLM-generated data against SMOTE and potentially CTGAN as traditional baselines. The goal is to figure out when (if ever) LLM synthetic data is a viable substitute or augmentation for real training data in classical ML pipelines.

Experimental Design

Models: Logistic Regression, SVM (RBF), XGBoost, MLP

Synthetic methods: LLM (Claude Sonnet 4.6), SMOTE, possibly CTGAN

Evaluation: Accuracy, F1, AUC-ROC on held-out real test set (80/20 split)

Seeds: 5 random seeds per condition, report mean +/- std

Full Data Regime (Substitution)

Total training size fixed at 5,000. We swap in synthetic data at different ratios while keeping the total constant. Tests whether replacing real data with synthetic hurts.

Condition	Real	Synthetic	Total
100/0 (baseline)	5,000	0	5,000
75/25	3,750	1,250	5,000
50/50	2,500	2,500	5,000
25/75	1,250	3,750	5,000
0/100	0	5,000	5,000

Low Data Regime (Augmentation)

Start with limited real data (roughly 10% of original or number of dimensions, whichever is larger). Then add synthetic data on top. Tests whether synthetic data helps when real data is scarce. Same ratios applied to the reduced pool.

Key Design Decisions

- Hyperparameters tuned once on real data only, then frozen across all conditions
- Test set is always 100% real, never touched during training
- Results CSV logs: dataset, regime, model, method, ratio, seed, accuracy, f1, auc
- Distribution constraints given in prompts to guide LLM generation

Dataset 1: Adult Income

Source: UCI ML Repository. 32,561 rows, 15 columns. Task: predict income >50K vs <=50K. Mix of categorical and numerical features (age, workclass, education, occupation, etc).

Generated 23,790 rows total across 51 batches (mix of 100-row and 500-row batches). Subsampled to 5,000 for experiments using random_state=42.

Prompt Design Notes

First batch had no distribution constraints. Results were bad: hours=40 appeared in only 2.7% of rows (real is 46.7%), Married-AF-spouse was overrepresented at 2.8% (real is 0.07%), and age mean was 47.2 (real is 38.6). Claude treated all categories as equally likely.

Added explicit distribution constraints for hours-per-week, age, marital-status, and workclass. Also added the full education-to-education-num mapping (16 levels). Quality improved dramatically in subsequent batches.

Validation Results (Combined 5,000 rows)

Exact matches with real data: 0

Duplicate rows: 0

Missing values: 0

Education-num mismatches: 0

Distribution Comparison

Metric	Real	Synthetic	Gap
Income >50K	24.1%	25.3%	+1.2%
Private workclass	69.7%	75.6%	+5.9%
Hours = 40	46.7%	~46%	~0%
Age mean	38.6	~38	~0
White race	85.4%	~86%	+0.6%

Note: distributions are close overall. The biggest gap is Private workclass at +5.9%, likely because we specified ~75% in the prompt but real data is actually 69.7%. Hours=40 was the biggest success story after adding distribution constraints.

Observations

- Without distribution constraints, Claude generates uniform-ish distributions across categories
- With constraints, marginal distributions are well-matched but joint distributions may still be off
- Claude likely has strong priors for this dataset (it's very well-known), which could inflate synthetic data quality
- Batch-to-batch variation exists (e.g., hours=40 ranged from ~44% to ~65% across batches) but averages out

Dataset 2: Default of Credit Card Clients

Source: UCI ML Repository. 30,000 rows, 24 columns. Task: predict whether customer defaults next month. More numerical features, sequential payment history (PAY_0 through PAY_6), and strong column dependencies (bills, payments, credit limits must be logically consistent).

Generated 5,000 rows across 10 batches of 500.

Schema Notes

This dataset is harder for LLMs because of inter-column dependencies. For example: payment amounts should generally not exceed bill amounts, bills should generally stay below the credit limit, and payment delay history (PAY_0 through PAY_6) should be temporally consistent. Real data also has PAY_X = -2 (about 9%), which wasn't in the original schema I drafted. Caught this by checking the actual data before finalizing the prompt.

Validation Results (Combined 5,000 rows)

Exact matches with real data: 0

Duplicate rows: 0

Missing values: 0

Invalid PAY_X values: 0

PAY_AMT > BILL_AMT violations: 3,029 (across 30,000 column-pairs)

BILL_AMT > LIMIT_BAL violations: 141

High delay but no default: 1,159

Distribution Comparison

Metric	Real	Synthetic	Gap
Default rate	22.1%	22.0%	-0.1%
Female (SEX=2)	60.4%	59.9%	-0.5%
University (EDU=2)	46.8%	46.6%	-0.2%
Age mean	35.5	35.1	-0.4
LIMIT_BAL mean	167,484	145,818	-12.9%
PAY_0 = 0	49.1%	43.0%	-6.1%

Note: LIMIT_BAL is 13% lower than real data, and PAY_0 distribution is off by 6%. These are the largest gaps across all three datasets. The logical consistency violations (PAY > BILL, BILL > LIMIT) are expected given that LLMs don't enforce arithmetic constraints natively. This is useful for the paper as it highlights where LLMs struggle.

Observations

- Marginal distributions (default rate, sex, education) are near-perfect
- Joint/conditional distributions are weaker: payment-bill-limit relationships aren't strictly maintained
- About 10% of payment-bill pairs violate the PAY_AMT <= BILL_AMT expectation

- This dataset will likely show the biggest performance gap between real and synthetic training
- The logical violations are actually a good finding: 'LLMs handle marginals well but struggle with dependencies'

Dataset 3: Heart Disease (Cleveland)

Source: UCI ML Repository (processed.cleveland.data). 303 rows, 14 columns. Task: predict presence of heart disease (binarized: 0 = no disease, 1 = disease). This is the smallest dataset and the one where synthetic augmentation should theoretically help the most.

Generated 5,000 rows across 10 batches of 500.

Schema Corrections

Initial schema had wrong category values for several columns. Caught by checking the actual file:

- cp is {1,2,3,4} in real data, NOT {0,1,2,3}
- slope is {1,2,3}, NOT {0,1,2}
- thal is {3,6,7}, NOT {1,2,3}
- Target is 0-4 in raw data, binarized to 0 vs 1+

Some Kaggle versions re-encode these to 0-indexed, but the raw UCI file uses original values. Would have generated invalid data without catching this.

Validation Results (Combined 5,000 rows)

Exact matches with real data: 0

Duplicate rows: 0

Missing values: 0

All category validations passed (cp, slope, thal, ca, etc): 0 violations

Distribution Comparison

Metric	Real	Synthetic	Gap
Disease rate	46.5%	44.3%	-2.2%
Male (sex=1)	68.0%	68.6%	+0.6%
cp = 4	47.5%	47.7%	+0.2%
Age mean	54.4	53.5	-0.9
Chol mean	246.7	253.9	+2.9%
Thalach mean	149.6	147.5	-1.4%
thal = 3	55.1%	55.7%	+0.6%

Distributions are tight across the board. The real question is whether Claude captured the subtle medical correlations (e.g., relationship between cp type and disease outcome, thal and ca interactions). Marginals look good but that doesn't guarantee the joint distribution is right.

Observations

- Best distribution match of all three datasets
- Real dataset is tiny (303 rows) so this is where low-data augmentation should shine
- Generated 5,000 rows from a 303-row dataset, so we're heavily extrapolating beyond the original data

- Medical domain knowledge in Claude's training probably helps here

Cross-Dataset Summary

	Adult Income	Credit Card	Heart Disease
Real rows	32,561	30,000	303
Synthetic rows	5,000	5,000	5,000
Columns	15	24	14
Feature types	Mixed cat/num	Mostly numerical	Mixed cat/num
Memorization	0 exact matches	0 exact matches	0 exact matches
Marginal quality	Good	Good	Very good
Logical consistency	Good	Weak (PAY/BILL)	Good
Key gap	Private wkclass +5.9%	LIMIT_BAL -12.9%	Disease rate -2.2%

Hypotheses Going Into Experiments

1. Adult Income will show the smallest real-vs-synthetic gap

Distributions are tight, and Claude almost certainly has strong priors for this dataset. Models trained on synthetic data should perform close to real-data baselines, especially for simpler models like logistic regression that mostly rely on marginal distributions.

2. Credit Card Default will show the largest gap

The inter-column dependencies (bill-payment-limit relationships) are the hardest for an LLM to get right, and we know from validation that ~10% of payment-bill pairs are logically violated. Tree-based models like XGBoost might be particularly affected since they can exploit these fine-grained feature interactions.

3. Heart Disease will benefit most from synthetic augmentation

With only 303 real rows, this is genuinely data-starved. Adding synthetic data in the low-data regime should help more here than for the other two datasets. The key question is whether the improvement comes from the extra data volume or from Claude capturing real medical correlations.

4. Simpler models may be more robust to synthetic data

Logistic regression primarily uses marginal feature-target correlations, which our synthetic data captures well. XGBoost and MLP can exploit higher-order interactions that the LLM may not preserve. So we'd expect logistic regression to degrade less on synthetic data than XGBoost.

5. SMOTE vs LLM comparison

SMOTE interpolates between existing real points, so it should preserve local structure better. LLM data is generated from a learned prior that may not reflect the specific dataset's structure. We expect SMOTE to outperform LLM data in the full-data regime (substitution), but LLM may do better in the low-data regime since SMOTE needs real neighbors to interpolate between.

Key Questions for the Analysis Phase

1. Does synthetic data help more on Heart Disease (tiny real set) than Adult Income (large real set)?
2. Does Credit Card Default show worse performance due to the financial logic violations?
3. Do simpler models (logistic regression) degrade more or less than complex ones (XGBoost) on synthetic data?
4. Is there a mixing ratio sweet spot? Maybe 75/25 real/synthetic is fine but 25/75 collapses?
5. In the low-data regime, does augmenting with LLM data actually help or just add noise?
6. How does LLM synthetic data compare to SMOTE? Is one consistently better, or does it depend on dataset/model?
7. Can we detect a memorization signal? Does Claude perform suspiciously well on Adult Income because it's seen the dataset?

Process Notes and Lessons Learned

Distribution constraints are essential. Without them, the first Adult Income batch had completely wrong distributions (hours=40 at 2.7% instead of 46.7%). Adding explicit frequency targets in the prompt fixed this almost entirely.

Always verify the schema against the actual data. Heart Disease had three columns (cp, slope, thal) with different value sets than what online descriptions said. Kaggle versions often re-encode columns, so the raw UCI file is the ground truth.

Batch-to-batch variation is real but averages out. Individual batches varied (e.g., hours=40 ranged from 44% to 65%) but the combined 5,000-row dataset was close to targets.

Claude maxes out around 500 rows per response regardless of what you ask for. Batches of 500 worked for most datasets, though some came back as ~100 rows.

Logical consistency is the weak point. LLMs can match marginal distributions when told to, but maintaining inter-column relationships (like `PAY_AMT <= BILL_AMT`) is hard. This is the most interesting finding so far and should be a focus of the analysis.

Validation scripts should be written before generation starts. Having automated checks caught issues immediately instead of finding them after generating thousands of rows.

Literature Context

Most prior work evaluates synthetic tabular data by training neural models on it, with limited attention to classical ML algorithms like logistic regression, SVMs, or tree-based methods. Existing studies also rarely compare LLM-based generation against established techniques (SMOTE, CTGAN) across varying data regimes and mixing ratios. Our project fills this gap.

Key references:

- GReaT (Borisov et al., 2023): foundational work on LLM-based tabular data generation
- CTGAN (Xu et al., 2019, NeurIPS): main traditional GAN baseline for tabular data
- SMOTE (Chawla et al., 2002): classic interpolation-based oversampling
- 'Are LLMs Naturally Good at Synthetic Tabular Data Generation?' (2024): critical view on LLM limitations
- 'Why do tree-based models still outperform deep learning on tabular data?' (Grinsztajn et al., NeurIPS 2022)